

THE UNIVERSITY OF DODOMA
COLLEGE OF INFORMATICS AND VIRTUAL EDUCATION
DEPARTMENT OF COMPUTER SCIENCE AND
ENGINEERING



UNDERGRADUATE UNIVERSITY
EXAMINATIONS

FIRST SEMESTER 2019/2020

CS 312: INTERNET PROGRAMMING AND
APPLICATIONS

Date: 17th February, 2020

Time Allocated: 3 Hours

Start: 11:45AM

End: 14:45PM

Instructions:

1. *This question paper consists of two sections: A and B.*
2. *Answer ALL questions in section A and ANY Two questions from section B*
3. *All University of Dodoma Examination Regulations Apply.*

SECTION A (60 Marks)

Question One

Choose the most correct answer and write its letter in the answer booklet: - **[1 Mark Each]**

- I. Which is the correct way to link to an external stylesheet from a web page
 - A. `<link rel = "stylesheet" type = "text/css" src = "styles.css" />`
 - B. `<link rel = "stylesheet" type = "text/css" href = "styles.css" />`
 - C. `<style href = "styles.css" type = "text/css" />`
 - D. `<style div{color: blue;} </style>`
- II. JavaScript is generally used as: -
 - A. a client-side scripting language
 - B. a server-side scripting language
 - C. an application development language
 - D. an embedded language for mobile devices
- III. Which JavaScript function is used to open a small window for user input: -
 - A. `alert()`
 - B. `prompt()`
 - C. `document.write()`
 - D. `init()`

- IV. Which property of a page element (eg a div) is used by JavaScript to alter the content of a page after it has loaded: -
- A. value
 - B. text
 - C. firstChild
 - D. innerHTML
- V. Which JavaScript object is used to get data from a server in an AJAX transaction: -
- A. Navigator
 - B. ajaxObject
 - C. XMLHttpRequest
 - D. none of the above
- VI. The JavaScript method used to obtain a reference to an element that has its id attribute set is: -
- A. getElementById()
 - B. element.getId()
 - C. document.getElementById()

D. `getElementsByTagName()`

VII. Which function is used to place the content of another file into a web page

- A. `place ()`
- B. `insert ()`
- C. `select ()`
- D. `include ()`

VIII. To maintain state in a web application values can be set in: -

- A. `$_SESSION`
- B. `$_REQUEST`
- C. `$_GLOBALS`
- D. `$_POST`

IX. A form has a text field: `<input type="text" name="phone" />`

When the form is submitted to a processing script, the text in this field can be found as the value of the variable: -

- A. `$_SESSION['name']`

- B. \$_POST['phone']
 - C. \$name
 - D. \$phone
- X. Which is not true of an XML document?
- A. It must have a root element
 - B. It must have a Schema approved by the W3C
 - C. All tags must be properly closed
 - D. The values of any attributes may be written in single quotes

Question Two

When dealing with files uploaded through HTTP, PHP stores references of them in the super global array. These files must be processed or moved from their temporary location during the lifetime of the PHP script execution or they will be automatically deleted. Write a PHP script to upload a file to a server. **[15 Marks]**

Question Three

Make a web page that uses a cookie to keep track of how many times a user has viewed the page. The first time a

particular user looks at the page, it should print something like "Number of views: 1." The second time the user looks at the page, it should print "Number of views: 2," and so on. **[10 Marks]**

Question Four

- I. Write a CSS rule that changes the color of all elements containing attribute *id*="Green-Move" to green and shift them down 25 pixels and right 15 pixels. **[5 Marks]**
- II. Write a layout template that contains two columns. Use *divs* for each layout component, and use float to line up the columns side by side. Give each component a border and/ or a background color so you can see where your *divs* are. **[5 Marks]**
- III. Several *divs* have a class of *even*. Write a style rule (as it would appear in an external stylesheet) that gives each of these black text on a red background using 14px sans-serif font, and the borders should be 1 pixel solid black lines all around. **[5 Marks]**
- IV. Write an SQL script to create a table called *users* in a database, with the following fields (assume the specific database has already been accessed)
id, which is the primary key and value is provided by the database server.

username, which has an appropriate string format.
password, with a format specific for the SHA encryption algorithm.

reg_date, with an appropriate date format.

[5 Marks]

- V. Which PHP functions are used to achieve the following with a MySQL database server on the local machine. **[2.5 Marks Each]**
- Connect to the server with username **root** and password **carrot2go**
 - Select a database called **ad006** to be used for queries

SECTION B (40 Marks)

Question Five

The University of Dodoma has asked you to create a program that allows prospective students to provide feedback about their campus visit. Your program should contain a form with text boxes for a name, address and email. Provide checkboxes that allow prospective students to indicate what they liked most about the campus. The checkboxes should include: student, location, campus, atmosphere, dorm rooms and sports. Also, provide radio buttons that ask the prospective student how they became

interested in the university. Option should include: friends, television, Internet and other. In addition, provide a text area for additional comments and Javascript function to validate each form field. When a student clicks submit button the program should insert those form fields' values in a table called *tbl_feedback* (name, address, email, yourLike, yourInterest, comments).

[20 Marks]

Question Six

Write a program that does basic arithmetic. Display a form with text box inputs for two operands and a <select> menu to choose an operation: addition, subtraction, multiplication, or division. Use a JavaScript function to validate the inputs to make sure that they are numeric and appropriate for the chosen operation. The processing function should display the operands, operator, and the result. For example, if the operands are 4 and 2 and the operation is multiplication, the processing function should display something like "4 * 2 = 8".

[20 Marks]

Question Seven

- I. Write a PHP script to get the PHP version and configuration information **[2 Marks]**
- II. What is the difference between echo and print? **[2 Marks]**
- III. Describe the differences between client/server interactions in traditional web applications and client/server interactions in Ajax web applications. **[5 Marks]**
- IV. Modules are helpful for many sites that include sections of repeated code. Explain the following three functions, make the differences between them clear:
include (), require (), require_once (). **[5 Marks]**
- V. Consider the following associative array

```
<? php
$ManU=array (
    "Rashford" =>"23",
    "Pogba" =>"24",
    "Degea" =>"22",
    "Martial" =>"25"
?>
```

- a. Write PHP code to add a new element with the key "Wan-Bisaka" and value "24". **[3 Marks]**
- b. Write PHP code to remove the element with the key "Pogba". **[3 Marks]**
